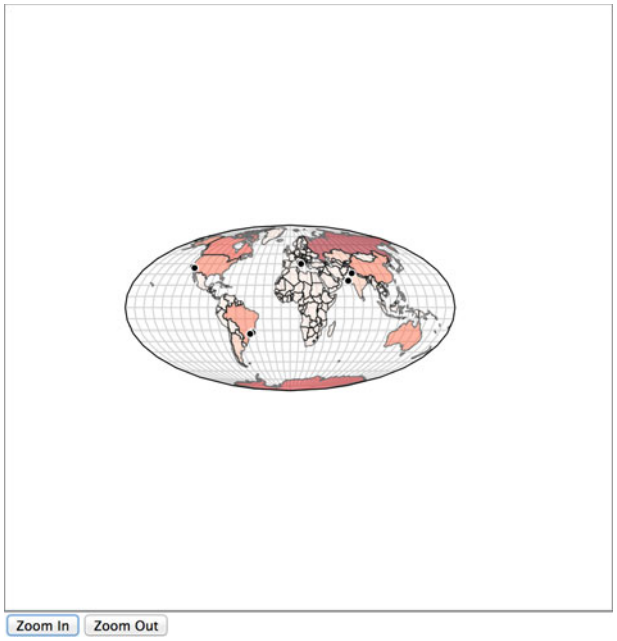


Figure 7.10 Zoom buttons and the effect of pressing Zoom Out five times. Because the zoom buttons modify the zoom behavior's translate and scale, any mouse interaction afterward reflects the updated settings.

The default zoom behavior assumes a user knows that the mousewheel and double-clicking are associated with zooming. But sometimes you want zoom buttons, because you can't assume the user knows that interaction or because you want to constrain or control the zooming process in a more complicated manner. The code in the following listing creates a zoom function and adds the necessary buttons, as seen in figure 7.10.

Listing 7.11 Manual zoom controls for maps

```
function zoomButton(zoomDirection) {
  if (zoomDirection == "in") {
    var newZoom = mapZoom.scale() * 1.5;
    var newX =
      ((mapZoom.translate()[0] - (width / 2)) * 1.5) + width / 2;
    var newY =
      ((mapZoom.translate()[1] - (height / 2)) * 1.5) + height / 2;
  }
  else if (zoomDirection == "out") {
    var newZoom = mapZoom.scale() * .75;
    var newX = ((mapZoom.translate()[0] - (width / 2)) * .75) + width / 2;
    var newY = ((mapZoom.translate()[1] - (height / 2)) * .75) + height / 2;
  }

  mapZoom.scale(newZoom).translate([newX,newY])
  zoomed();
}

d3.select("#controls").append("button").on("click", function (){
                      zoomButton("in")}).html("Zoom In");
```

Calculating the new scale is easy.

Calculating the new translate settings isn't so easy and requires that you recalculate the center.

Redraws the map based on the updated settings

Sets the zoom behavior's scale and translate settings to your new settings